
Yufeng (Alex) Xia

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PhD Candidate in Computer Science

EDUCATION

- **University of Edinburgh** Edinburgh, UK
MSc by Research in Computer Science, supervised by Prof. Mahesh K. Marina *Sep 2025 – Present*
- **University of Edinburgh** Edinburgh, UK
Bachelor of Science in Computer Science; Grade: First Class (Hons) *Sep 2021 – May 2025*

RESEARCH EXPERIENCE

- **University of Edinburgh** Edinburgh, UK
Junior Research Assistant *March 2024 – Present*
 - **Compiler Optimization Research:** Junior Research Assistant at Institute for Computing Systems Architecture (ICSA), under supervision of Dr. Antonio Barbalace.
 - **Embedded Systems:** Conducting research on compiler optimization techniques to improve performance and enable efficient code generation for embedded systems.
 - **Cross-disciplinary Projects:** Collaborating on cross-disciplinary projects involving compilers, and hardware-software co-design.
 - **Publications:** Authoring technical reports, presenting research findings at internal meetings, and preparing manuscripts for academic publications in the domain of compilers and system architectures.

RESEARCH AND PUBLICATIONS

- **Morphling: Emulator for Distributed Machine Learning at the Edge** EdgeSys 2026
Leyang Xue, Yufeng Xia, Eren Mendi, Ismaeel Bashir, Jiaxun Yang, Myungjin Lee, Mahesh Marina
 - **Scalable Edge ML Emulator:** Developed Morphling, a measurement-driven emulator that evaluates distributed ML training on heterogeneous edge devices at scale, supporting up to 1800 emulated devices on a single GPU server with minimal latency and thermal error.
- **Weaver: Foundation Model Training over AI-RAN Compute Infrastructure** MobiCom 2026
Leyang Xue, Tianxin Wang, Xin Zhe Khooi, Jiaxun Yang, Dheeraj Mahendiran, Yufeng Xia, Mun Choon Chan, Myungjin Lee, Mahesh Marina
 - **AI-RAN Infrastructure Sharing:** Developed Weaver, a system enabling opportunistic foundation model training on GPU-accelerated RAN infrastructure, achieving $2.1\text{--}4.2\times$ higher training throughput while harvesting up to 83% of spare compute without impacting RAN performance.
- **EmuRAN: Scalable and Cost-Effective Cloud-based RAN Emulation System** MobiCom 2026
Ujjwal Pawar, Andrew E. Ferguson, Yufeng Xia, Xenofon Foukas, Mahesh Marina, Bozidar Radunovic
 - **Scalable RAN Emulation:** Developed EmuRAN, a cloud-based RAN emulation system capable of emulating 10,000 mobile devices and 10,000 base stations on 130 cloud instances, two orders of magnitude larger than existing solutions.
- **CoSense: Auto-Quantization Based on Sensor Specifications for Compiler Optimizations** Under Review, 2025
YUFENG XIA, PEI MU, ANTONIO BARBALACE
 - **Framework Development:** Developed a novel LLVM-based auto-quantization framework leveraging sensor specifications to enhance performance and reduce code size for embedded systems.

AWARDS

- **Edinburgh Award:** Recognition of professional growth and invaluable experiences gained as a Research Assistant.